

The Cost of a Steeper Link: Calibration, Stability, and Tie Mechanics in Large-Scale Pairwise Preference Ranking

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Abstract

Large-scale preference leaderboards such as Chatbot Arena rank language models by maximum-likelihood Bradley–Terry (BT), inheriting the logistic link’s independence-of-irrelevant-alternatives assumption over a comparison pool that never stops growing. We test whether a flexible Thurstonian alternative—a lattice-based model that on pairwise data reduces to a steeper, natively tie-aware gap link—measurably improves refit stability, held-out calibration, or tie handling on 1.67 million Arena votes spanning 129 models and sixteen months. Every comparison was pre-registered: effect-size thresholds derived before fitting, synthetic detection gates passed before real data, verdict criteria fixed in advance. We find no demonstrable advantage in either direction. Refit stability under population growth is statistically indistinguishable (median paired rank-correlation difference $+0.00000$ across thirteen month-over-month refits); two structurally different tests agree that link shape is not discriminable at the ability gaps real matchups occupy (analytic ceilings of $0.23\text{--}0.31\times$ our minimum practical difference); and the calibration and tie-mechanism comparisons both end *inconclusive*, each dominated by a single-window episode rather than any systematic difference. Inside this null sit two structural findings: a steeper link’s single tie parameter entangles tie mass with decisive discrimination, whereas Davidson’s tie parameter is provably orthogonal to its logistic link; and under parameter uncertainty, steep links trade prediction-side overconfidence against an estimation-side shrinkage that once protected against a severe cold-start extrapolation error. One anomaly survives every attribution attempt: temporally localized excess dispersion under every link tested. Practically, cold-start coverage and nonstationary drift—not link choice—bind next-month predictability in deployments of this shape.

References

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